Foreground-only concurrency at streaming scale

How many viewers are actually watching, minute by minute — backgrounded and paused time excluded — computed entirely in ClickHouse, gated against raw events, and served in 7 ms.

2,917
ACTUALLY WATCHING AT THE PEAK
2026-07-26 10:56 · ANY-OVERLAP MINUTE RULE → S14

3,708
WHAT NAIVE SESSION-SPAN COUNTING SAYS
A 21.3% OVER-COUNT

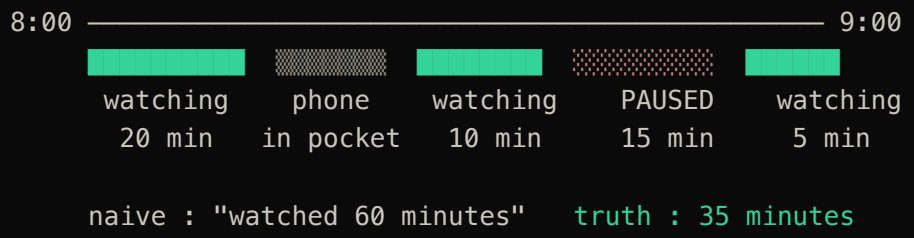
17,028
MINUTES RECONCILED AGAINST RAW
0 MISMATCHED

7 ms
HEADLINE CURVE, 329 KIB READ FROM SYSTEM.QUERY_LOG

> "How many people are watching right now?" — "the most-asked question in the building and one of the hardest to answer correctly."
>
> problem statement

> "Ingestion, modeling, and all concurrency computation live in ClickHouse." · "Apply foreground-only filtering." · "Publish continuously updated aggregates." · "Hand-computed answers score nothing."
>
> requirements · docs/upstream/, never edited

The app being open is **not** the same as someone watching:



THE FIVE QUESTIONS THE SPEC DEMANDS WE ANSWER WITH WORKING CODE

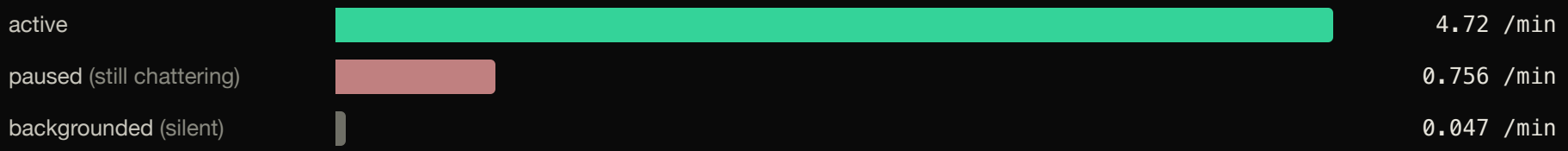
1	What is an **active interval** when the heartbeat is missing, the player paused, the app backgrounded?	slide 3
2	How are **active ranges represented** — intervals, deltas, hybrid?	slide 6
3	**Peak and average** without rescanning session history per query?	6 · 7
4	**Filter-friendly** across platform / country / content / type / grain?	5 · 11
5	**Still-open sessions** whose ranges keep growing?	slide 13

MEASURED ON THE PROVIDED FILE

33.6% of apparent watch time is backgrounded or paused. Foreground-only total: **1,978.1 h**. The whole problem exists to kill that over-count.

"Not watching" happens two ways — and only one is silent

PLAYER STATE	HEARTBEATS / MIN	MEDIAN WINDOW	DETECTABLE BY A GAP THRESHOLD?
actively watching	4.72	—	—
backgrounded	0.047	35 s	**YES** — a 100× drop; the pings stop
paused	0.756	20 s	**NO** — one ping every ~79 s slips under any sane threshold



THE HYBRID RULE — MEASURED BEFORE BUILT (ADR 0001, AMENDED BY ADR 0007)

```
active interval = contiguous run with no gap > 150 s   ← catches backgrounding
                MINUS explicit [pause, resume) windows  ← catches pause — gaps never can
                PLUS  60 s tail grace at a run ending in silence
```

A gap-only model would have shipped, passed every test we own, and silently booked paused time as watching — which the statement explicitly forbids. That is the difference between 3,708 and 2,917 at the peak.

raw events CSV 905,558 · sha256-pinned · tools/load.sh refuses to double-load

content CSV 33,464 titles → **content_dim** → dict_content (COMPLEX_KEY_HASHED — catalog plants a negative key)

▼

ev_raw · MergeTree · ORDER BY (toStartOfHour(ts), platform, video_session_id, ts) — the single source of truth; retains every event, so corrections need no stored state

▼ ▼

SESSION-AWARE (the accurate model)
session_intervals 30,323 · one row per contiguous *active* range · ReplacingMergeTree(build_version)
→ **cc_minute_delta** 28,073 · hour-clipped ±1 deltas · THE SERVING LAYER
→ **cc_hour_agg** 26,254 · peak + integral per hour, 8-level cube

SESSION-INDEPENDENT (the mandated baseline)
mv_stateless — a true streaming MV
→ **cc_minute_stateless** 91,292
cc_user_minute 91,692 · replaceable uniqExact buckets (ADR 0016 — MV retired)
peak 2,894 vs 2,917 at the same minute — the gap IS the excluded background+pause

▼ ▼

43 serving + health views — minute/hour/day, content-enriched, normalised, rolling + tumbling windows · a chart tool cannot read an AggregateFunction column; this layer is why it doesn't have to

ClickStack / HyperDX — the visualization reads the views AND our OTLP telemetry (slide 12) · zero custom UI code

THE GATE · /reconcile — recomputes truth from ev_raw with window functions (not the model's arraySplit), compares **every** minute incl. idle ones, exits 1 on any diff. 17,028 minutes · 0 mismatched · negative-tested twice (slide 8)

Keys follow the measured access shape — so our two tables sort in opposite directions

Our first key on `ev_raw` led with `video_session_id` "for session locality." The vendored ClickHouse best-practice rules said otherwise; we measured on the real 905K-event file, and the measurement won — over our own decision (ADR 0002).

ACCESS SHAPE	SESSION-FIRST KEY ROWS READ	HOUR-FIRST KEY ROWS READ	WIN
one-hour time slice	849,888	434,176	2.0×
hour + platform filter (the dashboard shape)	849,888	49,152	**17.3×**
full interval rebuild	905,558	905,558	identical

The locality argument was worth nothing: the rebuild touches **all** sessions, so it scans everything under either key — while every dashboard query filters by time and usually platform.

AND THE SERVING TABLES INVERT IT

`cc_minute_delta` · ORDER BY `(platform, country, content_id, minute, +4 dims)` — **dimensions first, time last**, because dashboards filter then range-scan: measured **122×** fewer rows read on the filtered A/B. Two opposite keys, one principle — and every CREATE TABLE carries a comment defending its ORDER BY, per docs/CONVENTIONS.md.

Active ranges become ±1 deltas, clipped at every hour boundary

A running sum over raw +1/−1 deltas needs every delta since the beginning of time — partition pruning becomes decorative. Clipping makes each hour's running sum **absolute and standalone**:

an interval running **20:59:48 → 22:04:49** emits:

```
UNCLIPPED   +1 @20:59 ····································· −1 @22:05
            to read 21:30 you must sum from t=0

CLIPPED     hour 20 | +1 @20:59   (survives the hour — no close emitted)
            hour 21 | +1 @21:00   (fresh open)
            hour 22 | +1 @22:00 … −1 @22:05
            every hour is now absolute — queries prune to the hours they touch
```

PAYOFF 1 — PRUNING IS REAL

Every running sum is `PARTITION BY toStartOfHour(minute)` — a repo-wide convention, because forgetting it yields *plausible wrong numbers*, not errors.

PAYOFF 2 — PEAK PRE-AGGREGATES ACROSS TIME

Per-hour `max` is a real number, not a fragment — so `cc_hour_agg` stores peak + integral per hour, and a day-grain peak reads **24 rows** per combination instead of 1,440 minutes.

Rejected representation: per-minute explosion of all history (148,900 rows vs ~28K — the "only works at hackathon size" choice the statement names). We keep it only as an independent cross-check for the gate, documented as not the serving path.

-State / -Merge, and the three arithmetic rules that keep the numbers honest

RULE	WHAT WE DO	MEASURED COST OF THE NAIVE WAY
Never sum a distinct count	cc_user_minute stores uniqExactState(user_id); reads uniqExactMerge; second hops use -MergeState. Session-level counts *are* summable across buckets (one session, one platform); user-level never.	**9×** over-count (45,000 vs truth 5,000)
Peak is not summable across dimensions	Always: filter → sum deltas per minute → running sum within hour → **then** max(). We never store a per-dimension peak; the 8-level cube computes each curve separately.	+2.0% (platform) **+53.6%** (content)
uniqExact, never uniq	HyperLogLog's 1–2% error is a silent correctness bug against an *exact* private answer key.	1–2% silent error

AND THE TYPE THAT IS LOAD-BEARING

`SimpleAggregateFunction(sum, Int64)` for delta columns. As `UInt64`, a negated corrective row (slide 13) silently wrapped to 2^{64}–n: `sum()` stayed right by modular arithmetic **so the bug hid**, while `max()` returned 1.8e19 and `starts` inflated 22%. Found by our own truncation test, fixed, re-gated.

Also enforced repo-wide: plain columns in an AggregatingMergeTree must be SimpleAggregateFunction (hard error on 26.7) · dictGet over joining the small dimension (3.7× less memory, measured) · reads may say FINAL, writers never say OPTIMIZE.

minutes_compared=17028 · mismatched=0 · max_abs_diff=0 · PASS

Independent by construction. `/reconcile` recomputes truth from `ev_raw` alone using **window functions** — a different algorithm from the model's `arraySplit` derivation — then compares a **dense minute spine**, so an idle minute can disagree too. A SUMMARY row asserts coverage; silence is failure, not success.

Negative-tested twice — a gate that cannot fail proves nothing:

`inject one bad delta row`	`exit 1`
`fabricate 500 viewers at an` **`idle`** `minute`	`25 mismatched · max_abs_diff 500`
`rebuild`	`exit 0`

And on every rebuild: delta layer vs interval expansion — 3,725 minutes, 0 mismatches · hour tier vs minute tier — 98 hours, 0 · incremental vs from-scratch — 1,578 minutes, 0.

THE STORY THAT MAKES IT CREDIBLE: WE BROKE OUR OWN GATE

An adversarial pass on the first gate found it **passing on zero rows**: five hard-coded 2026-07-26 timestamps meant any other day compared nothing, and `grep -q MISMATCH` read that silence as success. It was also blind to idle minutes — a fabricated 500 viewers at a quiet minute *passed*.

All three defects fixed and each fix negative-tested. Coverage went **5 → 17,028 minutes**, idle minutes included, on the same zero-mismatch verdict — and the gate now derives its target minutes from the data, so it works unchanged on the unseen day (proven on a holdout: 1,364 minutes, 207 of them idle).

What the gate cannot see — a definitional error, because it recomputes truth from the same spec it is testing. That honest boundary is slide 14.

Every "best practice" was made to prove itself on our data. Six didn't.

CANDIDATE	LOOKED LIKE	MEASURED, ON THE REAL FILE
PROJECTION on ev_raw by session id	27.7× on single-session lookups	the straggler path of the day used IN (subquery), which full-scans → **1.00×**. Re-measured when ADR 0013 changed the access pattern: **12.8×** on the finalizer shape, still +91% storage. **Not shipped either way** — a measurement is only valid for the shape it was taken on
argMax() instead of FINAL	"FINAL is slow" folklore	**3–4× slower, 5.7× more bytes** — one active part, zero duplicate keys: FINAL has nothing to collapse
a dedup step (spec step 3)	mandatory hygiene	4,210 duplicate rows are **provably inert**: raw vs deduped derivation identical — **0 of 3,725 minutes differ** (0 of 834 restricted to the 863 duplicate-bearing sessions). "We proved the step unnecessary" beats "we added the step"
any() to attribute dimensions	a cheap tie-break	picks the player's pre-resolution **sentinel**: **44.5%** of audio attributions wrong, 73.5% at the peak minute — and non-deterministic across thread counts. Replaced by dominant-value-per-interval → 3.3%
uniq() (HyperLogLog)	faster distinct	1–2% error against an exact private key — a silent correctness bug
app-side insert batching	standard ingest advice	async_insert=1 already does it server-side; we would reimplement the server and move durability into a process that can die holding data

Not one decision was settled by argument — and **four measurements overturned our own prior plan**, twice reversing a choice we had already ADR'd. The defensible story is not "we designed it well"; it is "we kept trying to disprove it, and here is the list of times we succeeded."

We quote what queries read, not just how fast they return

7 ms	329 KiB	28,073	8.5×
HEADLINE CONCURRENCY CURVE QUERY CACHE OFF	BYTES READ FOR THAT CURVE	ROWS READ — THE WHOLE DELTA LAYER	LESS I/O THAN EXPANDING INTERVALS 299 KB VS 2.55 MB

Evidence, not vibes. Every benchmark run is tagged `SETTINGS log_comment='…'` and the numbers are read back from `system.query_log` — the authoritative record of latency, rows and bytes. No screenshot, no stopwatch, reproducible by anyone with the service.

A filtered dashboard shape (`ck1_android_phone`): **30 ms · 603 KiB** — still reading the 28K-row delta layer, never raw history.

Why it's cheap by construction: queries read `cc_minute_delta` — rows exist only where concurrency *changes*, hour-clipping bounds every scan to the hours touched, and the dimension-first sort key (slide 5) prunes the rest.

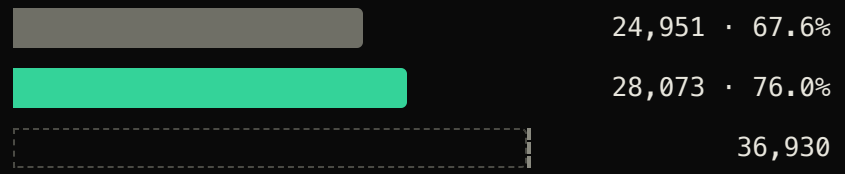
And it's observable: a HyperDX source charts `system.query_log` itself — p95 latency and bytes read of our own queries, on the graded service.

Window views use RANGE frames, not ROWS — a delta table stores only change rows, so "5 rows back" is not "5 minutes back". Rolling/tumbling views verified against a brute-force join: 0 mismatches.

> "The provided dataset is a scaled-down proxy for a petabyte-class production problem. Judges will ask how your design behaves at 100×."
>
> problem statement — scale framing

THE HARD ROW CEILING — A BOUND, NOT A HOPE

`cc_minute_delta` stores at most one open + one close per (merged run, hour), so on this file it is capped at **≤ 36,930 rows no matter how many dimensions exist**:



Whole table: 111 KiB. Per-minute explosion (148,900 rows, ≈5.3× larger) has no such ceiling — that is why it isn't the serving path.

Work scales with **change**, not history: the delta layer is O(intervals), rebuilds touch only sessions that changed, corrections cost O(stragglers). Hour-clipping means no query ever scans from t=0.

WHAT BREAKS FIRST — SAID BEFORE WE'RE ASKED

The graded DB is batch-published today. The incremental publisher (ADR 0013/0016) is proven byte-identical to a rebuild across all four tiers in scratch — but it has never committed a run on the graded DB, and its crash/concurrency windows are audited by inspection only (slides 13 · 15).

FINAL vs argMax flips at scale. Our measurement (slide 9) holds at one active part and zero duplicate keys; with many parts and frequent corrections it reverses. We would re-measure, not assume.

The file can't test everything. country has a single value, so a country-filter bug is invisible to every test we own — we test on platform's 10 values instead, and say so.

ClickStack charts our data and observes our pipeline. Delete it and the demo breaks.

DIRECTION 1 — CLICKSTACK IS THE VISUALIZATION

HyperDX sources read our serving views on the graded Cloud service: the concurrency curve, per-dimension drilldowns, content-level views — and a `system.query_log` source charting the p95 latency and bytes read of our own queries. **Zero custom UI code**: the spec puts polished frontends out of scope and separately requires an OSS integration — one component satisfies both.

DIRECTION 2 — CLICKSTACK OBSERVES THE PIPELINE

`sonyliv observe` (Go) emits over OTLP into ClickStack's `otel_*` tables: **watermark lag** — the observable expression of the whole update design — plus build-stage timing and the reconcile gate verdict, rendered as a freshness panel. A ~250-line stdlib-only OTLP/HTTP JSON client; no SDK.

The test we set ourselves — the organiser warns "superficial inclusion won't count": remove ClickStack and the freshness panel has *nowhere to come from*. The integration is load-bearing in both directions, not a logo on a slide.

OSS depth beyond ClickStack: the official `ClickHouse/agent-skills` best-practice rules (Apache-2.0) are vendored into the repo and cited by name in our ADRs — and they **overturned one of our own schema choices** (the ev_raw sort key, slide 5). We treat upstream OSS guidance as a reviewer, not decoration.

Go CLI: verify (env/schema preflight) · observe (OTLP emit). It computes nothing about concurrency — R1 stays intact: all modeling and computation live in ClickHouse.

Nothing is ever mutated. Every late arrival is re-derived per session and appended as a diff — never rebuilt.

ARRIVAL CLASS	MECHANISM	WHY IT IS CORRECT
in order, near-real-time	an MV on ev_raw records touched sessions; the **publisher** claims what arrived since its cursor and re-derives only those (ADR 0013)	appends the negation of each session's published deltas plus its new ones; cost scales with changed sessions, not history
straggler, however late (measured tail: 2,081 s after VideoSessionEnd)	**correction-by-diff** (ADR 0006, live): recompute that one session, append −old +new	exactly as correct as a full rebuild — because it *is* one, of one session. SimpleAggregateFunction(sum, Int64) absorbs the negative rows natively; a 46-min straggler measured at 3.4 s
hour/day peaks + distinct users	**publisher phases** re-derive every touched bucket; cc_user_minute is replaceable buckets, not a set union (ADR 0016)	replacement is the cheapest representation in which user *retraction* exists — all four tiers converge to a from-scratch rebuild: 0 differing cells across bootstrap, shrink, dimension flip, straggler, 200 forced runs

THE PROOF — A TRUNCATION TEST THAT MANUFACTURES THE WORST CASE

Cut the stream at the peak minute — **52.6% of sessions left open** — then absorb the withheld **447,081 late events** incrementally: never truncating, never mutating, only appending. Result: row-for-row equal to a from-scratch build on all **1,578 minutes, 0 mismatches**.

It caught **two real bugs** before any judge could: a stale-row defect (versioning intervals by `interval_end` let an overshot tail-grace win forever — 316 intervals, peak over-counted by 37; now `ReplacingMergeTree(build_version)`) and the UInt64 wrap on negated corrections (slide 7). Both fixed, both re-gated.

Honest status: the publisher is proven in a scratch database (evidence/publish.txt); the graded DB is still batch-rebuilt — the installed publisher has never committed a run there (slide 15). The hot tier (ADR 0004/0005) was **retired, not deferred**: correction-by-diff makes it unnecessary, and its lease model counted paused time as watching.

Five definitions the data cannot settle — each measured, each built as a switch

FORK	OUR DEFAULT, STATED	MEASURED WORTH
1 · minute membership. Is a session concurrent at minute M if it overlaps *any part* of M, or only if active *at the instant M:00* — how a sampled gauge reads? The gate cannot see this: it shares the convention.	any overlap counts — consistent across model, gate and serving tiers	**2,507 vs 2,917 · −14.1%** the largest fork measured (doubts/09)
2 · resume semantics. 9,958 resume→resume pairs; 900 sessions whose first pause/resume event is a *resume*. Does a pause close at the first resume, or one that corresponds to it?	close at first resume; unpaired resumes are noise	**189.2 h · 9.6%** of counted watch time
3 · unclosed pauses. 23% of pauses never resume. Stay paused to end of run, or end at next event?	conservative — stays paused (over-counting invents viewers)	**99.3 h · +131 peak** pre-ADR-0009 measurement; permissive arm re-run pending
4 · heartbeat cadence. The doc says 60 s; the shipped data ticks at **40.0 s** (p50 = p90 on three separate telemetry streams). Which did the ground truth assume?	the shipped data is authoritative	**≤ 170.9 h · 8.6%** upper bound
5 · is Hindi one language or four? hin / HIN / hin-hindi / hin-Hindi are four raw values.	store raw, normalise on read — never degrade the derivation	**+23.8%** on any per-language answer (peak 1,774 → 2,196, live)

None of these is measurable from the data: the ground truth is private, so a wrong guess is silently wrong on every benchmark answer. Our gate **structurally cannot catch them** — it recomputes truth from the same rule. That is why each fork is a **one-line switch** (or a stated two-file convention) with an ~11 s rebuild: the moment a definition is pinned, we re-run and re-gate.

Also disclosed: `country` carries a single value in the provided file, so country-filter bugs are invisible to our tests — we validate filtering on platform's 10 values and say so out loud.

`doubts/` — one dossier per fork: evidence, exact wording, a decision table per answer

naming the limit pre-empts the question — this project's story is "we kept trying to disprove ourselves"

Next steps, in scoring order — each already de-risked

STEP	STATE OF DE-RISKING
1 · First committed publisher run on the graded DB — the spec's "continuously updated aggregates"	the mechanism is no longer a gap: the incremental finalizer maintains **all four tiers** (ADR 0013/0016), proven byte-identical to a from-scratch rebuild — **0 differing cells** across bootstrap, growth, shrink, dimension flip, a 46-min straggler and 200 forced runs; phases 0.5–1.8 s. The graded DB has the schema installed and zero committed runs
2 · Live continuous-publication demo — publisher on a per-minute cadence, watermark lag charted in ClickStack	every piece exists: publish batches measured 3.5–5 s in scratch, sonyliv observe already emits the lag. (The hot tier was retired, not deferred — ADR 0013 measured correction-by-diff making it unnecessary)
3 · Close the publisher's crash / concurrency windows	consumed-before-claimed and single-publisher assumptions are audited by inspection only — reproduce, then guard (queued as ADR 0017)
4 · Commit the tail-sensitivity sweep as a harness	already measured by the cross-model audit: gap {120, 150, 180} × tail {0, 60, 150} spans peak **2,756 → 3,132** (13.6%); our shipped point 150/60 → 2,917. Make it one command so the unseen day re-runs it
5 · /bench on the official query set, if released	it never was — a **13-query reconstruction** of the statement's shapes is committed with answers, bytes read and query ids (evidence/benchmark/): largest read 814 KiB, server medians 7–45 ms, ev_raw never touched
6 · Ingest definition answers (slide 14)	every fork is a one-line switch (minute membership: a stated two-file convention) and an ~11 s rebuild, re-gated by /reconcile

THE CLOSE

Correct first — 17,028 minutes gated against raw events, negative-tested. **Fast second** — 7 ms and 329 KiB for the headline curve, and we quote what queries *read*. **Update-friendly third** — one mechanism for every arrival class, all four tiers converging on the rebuild answer, proven by tests that caught two real bugs. And where a definition is genuinely open, we measured what it is worth and built the switch.